CHEMICAL EQUILIBRIUM

(a) Irreversible Reaction:

A unidirectional reaction in which reactants (at least the limiting reactant) is completely consumed and reaction goes to completion in a definite time period.

eg. of irreversible reaction,

→ Acid - Base Neutralisation rxn.

$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

→ Precipitation reaction.

$AgNO_3(aq) + KCl(aq) \longrightarrow \underset{\text{white ppt.}}{AgCl(\downarrow)} + KNO_3(aq)$

→ Thermal decomposition of $CaCO_3(s)$ in open vessel

$CaCO_3(s) \xrightarrow[\Delta]{} CaO(s) + CO_2(g)\uparrow$

[In open vessel]

(b) Reversible Reaction:

A bidirectional reaction in which none of the reactant is ~~completly~~ completely consumed and a State called equilibrium state is reached. at which rate of both forward and backward reaction become equal.

→ At equilibrium state, concentration or partial pressure of the species involved in reaction become constant. Also all thermodynamic parameters like ΔH, ΔS, ΔG etc also become constant. Also colour, density

and other measurable properties at equilibrium become constant.

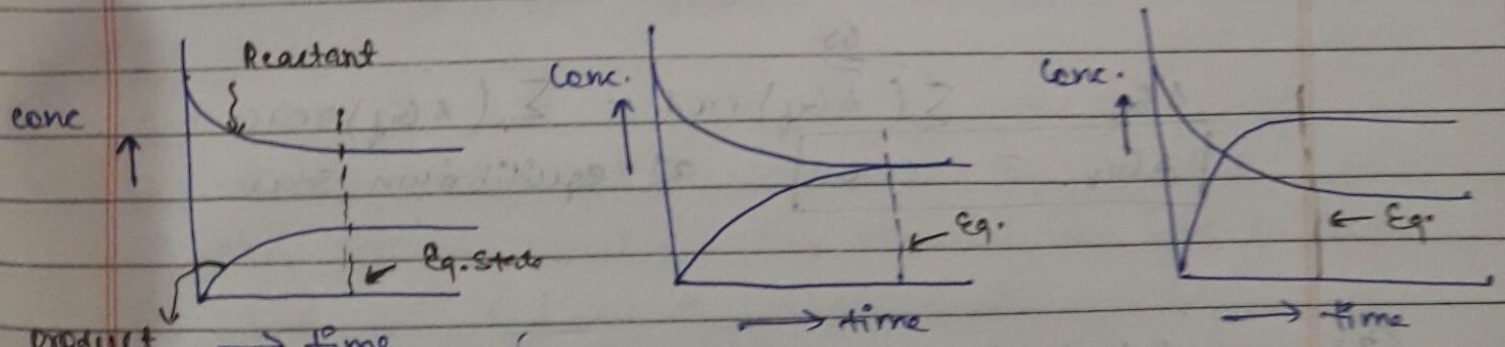
* A true eqm attains in closed system

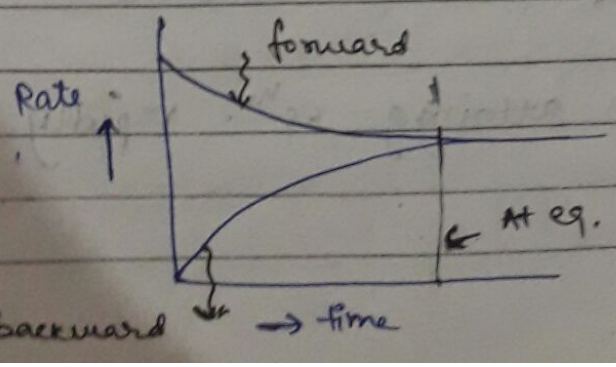
eg. of reversible reaction

- $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$
- $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$
- Haber's Process

  $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
- $H_2O(s) \rightleftharpoons H_2O(l)$
- $C(diamond) \rightleftharpoons C(graphite)$
- $CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$

(*) Graph for reversible reaction :-

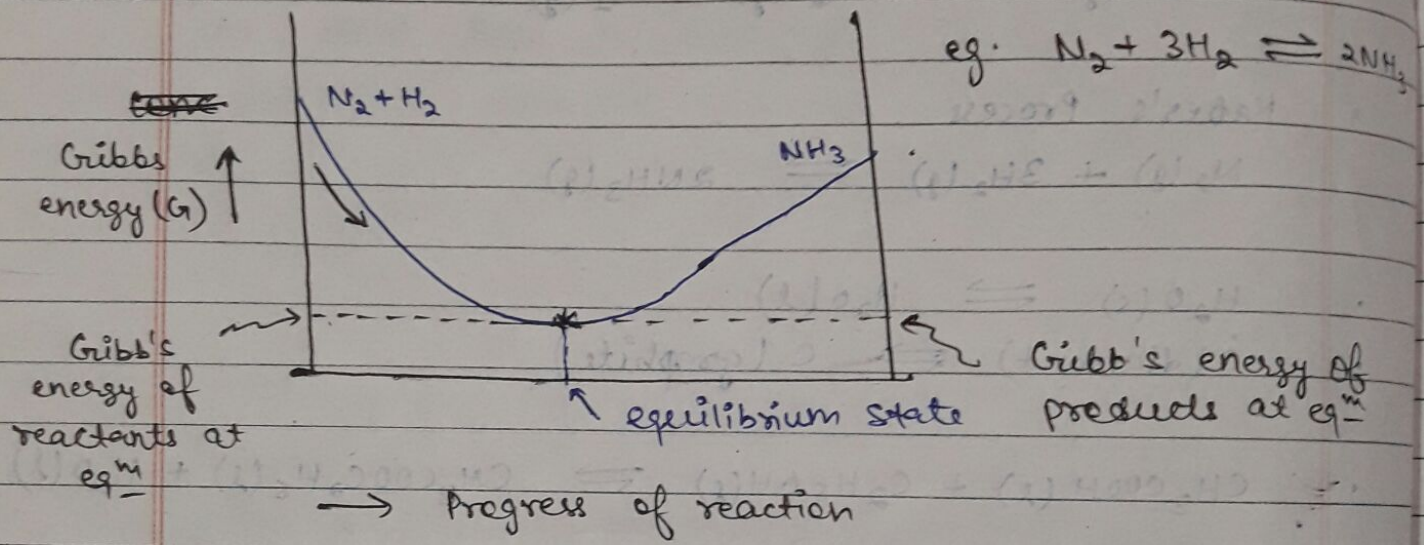




$R_f = R_b$ at equ.

Imp * Characteristics or properties of Chemical equilibrium :-

1. Approach of equilibrium is always spontaneous
Equilibrium state is the most stable state for any system and hence equilibrium will arrive by moving reaction in forward or backward direction.



→ At eqm, Gibb's energy at constant pressure & temperature become minimum.

$$\Delta G_r = \sum (\Delta G_f)_{prod.} - \sum (\Delta G_f)_{react.}$$

or

$$\boxed{\Delta G_r = 0}$$ at equilibrium state.

(2.) Catalyst does not affect the position of eqm. and also the value of equilibrium constant

→ Catalyst helps attaining eqm. rapidly

3. Nature of eqm is dynamic, that means at eqm. state both forward and backward reaction proceed with equal rate.

$R_f = R_b$ at eqm

4. Concentration or partial pressure of all the species involved at eqm state remains constant.

⊛ Types of equilibrium :

(a) Homogeneous eqm :

When all the reactants and products involved in reaction are in same phase.

eg. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

$N_{2(g)} + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$CH_3COOC_2H_5(aq) + H_2O(l) \rightleftharpoons CH_3COOH(aq) + C_2H_5OH(aq)$

(b) Heterogeneous eqm :

When all the reactants and products are not in same phase.

eg. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

$S(s) \rightleftharpoons S(l)$

$C_{(graphite, s)} \rightleftharpoons C_{(diamond, s)}$

* **Phase :** matter which has uniform composition and uniform properties throughout.

→ A same physical state of a substance may have two different phases.

eg. • C(dia.) and C (graphite) are two different phases.

• S(rhombic), S(mono) are two different phases.

• P(white), P(Red), P(black) are all solids but different phases.

Classification of Heterogeneous equillibrium

(i) Phase Change eqm :

When same substance undergoes phase change eg. (allotropic change or Physical state change)

$H_2O(s) \rightleftharpoons H_2O(l)$

$H_2O(l) \rightleftharpoons H_2O(g)$

(ii) $S(s) \rightleftharpoons S(l)$

(ii) Heterogeneous chemical eqm :

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

$C(s) + CO_2(g) \rightleftharpoons 2CO_2(g)$

* Law of mass action :
(Given by Guldberg and Waage)

At a given temperature, Rate of a reaction is directly proportional to the active masses of reactants raised to power their stoichiometric coefficient.

eg. $aA + bB \rightleftharpoons cC + dD$

→ Active mass of a substance are most commonly taken as molar concentration or partial pressure.

Rate of forward reaction,

$$R_f \propto [A]^a \times [B]^b$$
$$= K_f [A]^a \times [B]^b \quad \left[K_f = \text{Rate constant of forward rxn.} \right]$$

Similarly, for Backward reaction,

$$R_b \propto [C]^c \times [D]^d$$
$$R_b = K_b [C]^c [D]^d \quad \left[K_b = \text{Rate constant of backward rxn.} \right]$$

OR

Specific rxn. Rate.

* K_f, K_b → independent of conc. or partial pressure
→ depend on temperature and catalyst.

(*) Active mass of various substances.

1. For substance in solution phase, active mass will be molar concentration.

eg. $I_2(aq) + I^-(aq) \underset{K_b}{\overset{K_f}{\rightleftharpoons}} I_3^-(aq)$

$$R_f = K_f \times [I_2][I^-]$$
$$R_b = K_b \times [I_3^-]$$

(2) For gaseous substance active mass may be taken as molar conc. or partial pressure.

eg. $N_2(g) + 3H_2(g) \underset{k_b}{\overset{k_f}{\rightleftharpoons}} 2NH_3(g)$

$R_f = k_f \times [N_2][H_2]^3$
$R_b = K_b \times [NH_3]^2$
} in terms of Molar concentration

$R_f = k_f' \times P_{N_2} \times P_{H_2}^3$
$R_b = k_b' \times P_{NH_3}^2$
} in terms of partial pressure.

(3) Active mass of pure solid and pure liquid is constant and does not depend on amount of substance

eg. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

$[CaCO_3(s)]$ ← active mass of $CaCO_3$

$= \frac{n_{CaCO_3}}{V_{CaCO_3}} = \frac{W_{CaCO_3}}{M_{CaCO_3} \times V_{CaCO_3}}$

$= \frac{d_{CaCO_3}}{M_{CaCO_3}} = \text{constant}$

$R_f = k_f \times [CaCO_3] = k_f'$ (Modified rate constant of forward rxn.)

$R_b = k_b \times [CaO_{(s)}][CO_2]$
$= k_b' [CO_2]$

* In the expression of Rate of reaction and equilibrium constant (K_p or K_c), active mass of pure solid or pure liquid is teeken 1.

(4) Active mass of Solvent in excess is also constant and hence taken 1 in expression of Rate of rkn. and Equilibrium constant.

eg. $CH_3COOC_2H_5(aq) + \underset{(solvent)}{H_2O(l)} \rightleftharpoons CH_3COOH(aq) + C_2H_5OH(aq)$

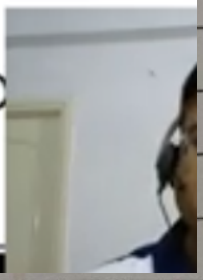
$$R_f = K_f[CH_3COOC_2H_5] \times 1$$
$$R_b = K_b[CH_3COOH][C_2H_5OH]$$

Note :- Agar H_2O Reactant ke side hai to constant Hoga aur product ke side constant nahi Hoga.

not a solvet

* $CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$

$$R_b = K_b[CH_3COOC_2H_5][H_2O]$$



* Law of Chemical Equilibrium :

At eq^m state,

Rate of forward rxn. (R_f) = Rate of backward rxn. (R_b)

eg. $aA + bB \rightleftharpoons cC + dD$

$$R_f = K_f \times [A]^a [B]^b$$

$$R_b = K_b \times [C]^c [D]^d$$

At eq^m, $R_f = R_b$

$$K_f [A]^a_{eq} \times [B]^b_{eq} = K_b \times [C]^c_{eq} \times [D]^d_{eq}$$

$$\Rightarrow \frac{K_f}{K_b} = \frac{[C]^c_{eq} \times [D]^d_{eq}}{[A]^a_{eq} \times [B]^b_{eq}} = K_{eq}^m$$

K_{eq} (or Equilibrium constant)

→ K_{eq}^m does not depend on
(i) Conc. or partial pressure
(ii) Presence of Catalyst
(iii) ~~Catalyst~~ Inert Gas

→ K_{eq} for a reaction depends on temperature.

* Types of equilibrium constant :

(1.) K_c : → When active masses are taken as Molar concentration.

→ K_c is defined for gaseous rxn. as well as reactions occuring in solution phase.

eg. • $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$$K_c = \frac{[NH_3]^2_{eq}}{[N_2]_{eq} \times [H_2]^3_{eq}}$$

• $CH_3COOC_2H_5(aq) + \underset{\text{(solvent)}}{H_2O(l)} \rightleftharpoons CH_3COOH(aq) + C_2H_5OH(aq)$

$$K_c = \frac{[CH_3COOH][C_2H_5OH]}{[CH_3COOC_2H_5]}$$

(2) K_p : → Can be written for a gaseous rxn. with/without having pure solid or pure liquid.

eg. • $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$$K_p = \frac{P^2_{NH_3}}{P_{N_2} \times P^3_{H_2}}$$

• $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

$$K_p = (P_{CO_2})_{eq}$$

③ K_{pc} : → for rxn involving substance in solution phase and gas phase as well.

eg. $H_{2(g)} + Cl_2(g) \rightleftharpoons 2H^+(aq) + 2Cl^-(aq)$

$$K_{pc} = \frac{[H^+]^2 \times [Cl^-]^2}{P_{H_2} \times P_{Cl_2}}$$

* Unit of eqm constant :

eg. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$$ unit : $(mol/lit)^{-2}$ (OR) $(mol/lit)^{\Delta ng}$

$$K_p = \frac{P_{NH_3}^2}{P_{N_2} \times P_{H_2}^3}$$ unit : $(atm)^{-2}$ or $(Pa)^{-2}$ or $(atm)^{\Delta ng}$

* Dimensionless equilibrium Constant :

→ All conc. or partial pressure are taken relative to Standard conc. or Standard partial pressure.

Standard Conc. : 1 mol/lit or 1M
Standard Partial pr : 1 bar

eg. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

$$K_c^\circ = \frac{\frac{[PCl_3]}{[PCl_3]_\circ} \times \frac{[Cl_2]}{[Cl_2]_\circ}}{\frac{[PCl_5]}{[PCl_5]_\circ}}$$

$[PCl_5]_\circ = 1M$
$[PCl_3]_\circ = 1M$
$[Cl_2]_\circ = 1M$

unitless or dimensionless term

$\therefore \Rightarrow$ Numerically K_c° and K_c are same

$$K_p^\circ = \frac{\frac{P_{PCl_3}}{P^\circ_{PCl_3}} \times \frac{P_{Cl_2}}{P^\circ_{Cl_2}}}{\frac{P_{PCl_5}}{P^\circ_{PCl_5}}}$$

$P^\circ PCl_5 = P^\circ PCl_3 = P^\circ Cl_2 = 1$ bar

dimensionless term

$K_p^\circ = K_p \times (1.013)^{\Delta n_g}$

when all partial pr. of gases are taken in atm.

$\longrightarrow$ We take $K_p^\circ \cong K_p$ when pr are taken in atm.

(*) Relation between K_p and K_c for a gaseous reaction —

eg. $aA_{(g)} + bB_{(g)} \rightleftharpoons cC_{(g)} + dD_{(g)}$

$$K_p = \frac{P_C^c \times P_D^d}{P_A^a \times P_B^b}, \quad K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$\Rightarrow$ for a ideal gas, $PV = nRT$

$P = n/V \times RT$

$P = \text{Conc.} \times RT$

$\therefore P_A = [A] \times RT$

$P_B = [B] \times RT$

Now, $K_p = \frac{[C]^c \times (RT)^c \times [D]^d \times (RT)^d}{[A]^a \times (RT)^a \times [B]^b \times (RT)^b}$

$$K_p = K_c (RT)^{c+d-a-b}$$

$$\Rightarrow \boxed{K_p = K_c (RT)^{\Delta ng}}$$

where Δng = Σ of stoichiometric coeff. of gas product − Σ of st. coeff. of gas reactant

eg. • $C_{(s)} + \frac{1}{2} O_{2(g)} \rightleftharpoons CO_{(g)}$, $\Delta ng = 1 - \frac{1}{2} = \frac{1}{2}$

• $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$, $\Delta ng = 2 - 4 = -2$

* Characteristics / Properties of Equilibrium Constant (K_{eq})

1. $A \rightleftharpoons B$, $K_c = \frac{[B]}{[A]}$

Rxn is reversed

$B \rightleftharpoons A$, $K_c' = \frac{[A]}{[B]} = \frac{1}{K_c}$

2. $X \rightleftharpoons Y$, $K_p = \frac{P_Y}{P_X}$

Multiply the eqn. by 2

$2X \rightleftharpoons 2Y$, $K_p' = \frac{P_Y^2}{P_X^2} = (K_p)^2$

3. $A + B \rightleftharpoons C + D$ $\quad$ $X \rightleftharpoons Y$

$$K_{c_1} = \frac{[C][D]}{[A][B]} \qquad K_{c_2} = \frac{[Y]}{[X]}$$

on adding

$$A + B + X \rightleftharpoons C + D + Y$$

$$K_c = \frac{[C][D][Y]}{[A][B][X]}$$

$$\boxed{K_c = K_{c_1} \times K_{c_2}}$$

4. Effect of temperature on Keq:

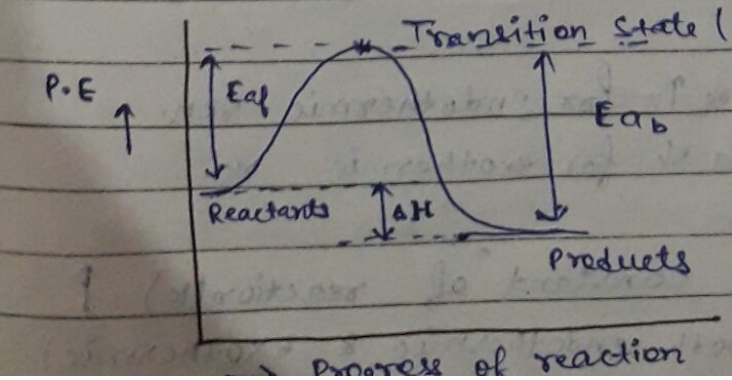
Using the Arrhenius eqn.

Rate constant, $k = Ae^{-E_a/RT}$

where A = ~~frequency~~ factor or Arrhenius constant
(Independent of temperature (constant))

E_a = Activation Energy of reaction.

T = Temperature (in Kelvin)



$$\Delta H_r = E_{af} - E_{ab}$$

Activation Energy of forward reaction

Activation Energy of Backward reaction.

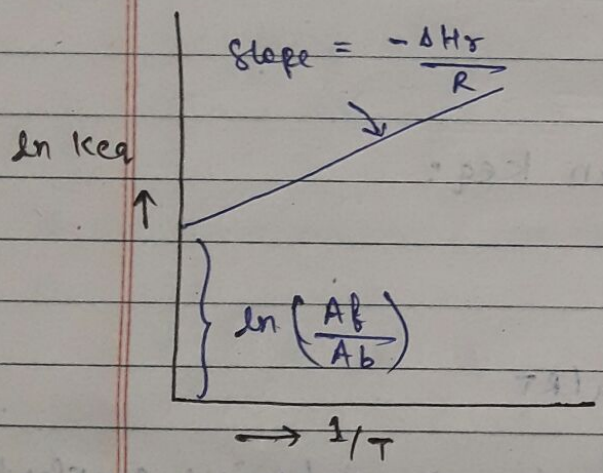
$$K_{eq} = \frac{k_f}{k_b} = \frac{A_f e^{-E_{af}/RT}}{A_b e^{-E_{ab}/RT}}$$

$$K_{eq} = \frac{A_f}{A_b} e^{-(E_{af}-E_{ab})/RT}$$

$$K_{eq} = \frac{A_f}{A_b} e^{-\Delta H_r/RT}$$

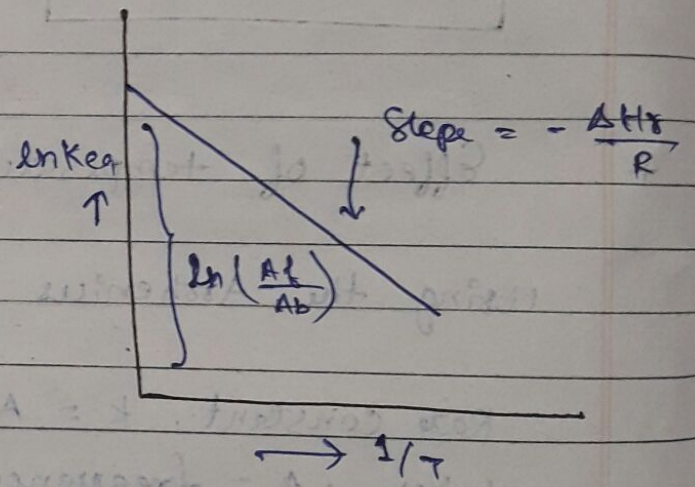
$$\ln K_{eq} = \ln \frac{A_f}{A_b} - \frac{\Delta H_r}{RT} \times (-1) \quad \text{---(i)}$$

$$y = mx + c$$



For exothermic rkn.
(ΔH_r = -ve)

T↓, Keq ↑
T↑, Keq ↓



For endothermic rkn.
(ΔH_r = +ve)

T↓, Keq ↓
T↑, Keq ↑

* NOTE : T↑, Keq ↑ for endothermic rkn.
T↑, Keq ↓ for exothermic rkn.

But, T↑, Rate of constant of reaction (k) ↑
(for both endothermic & exothermic) rkn.

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Now,

$$\ln K_{eq} = \ln \frac{A_f}{A_b} - \frac{\Delta H_r}{RT} \quad \text{— (i)}$$

At temp. T_1, $\ln K_{eq_1} = \ln \frac{A_f}{A_b} - \frac{\Delta H_r}{RT_1}$

At temp. T_2, $\ln K_{eq_2} = \ln \frac{A_f}{A_b} - \frac{\Delta H_r}{RT_2}$

$$\ln \frac{K_{eq_2}}{K_{eq_1}} = \frac{\Delta H_r}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

Vant Hoff Equation.

$$\text{Or } \log_{10} \frac{K_{eq_2}}{K_{eq_1}} = \frac{\Delta H_r}{2.303\,R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

(*) Applications of equilibrium constant :

1. Prediction of direction of reaction.

eg. $a\underset{(g)}{A} + b\underset{(g)}{B} \rightleftharpoons c\underset{(g)}{C} + d\underset{(g)}{D}$,

At any instant, a term called reaction quotient can be calculated as

$$Q_c = \frac{[C]_t^c \times [D]_t^d}{[A]_t^a \times [B]_t^b} \quad ; \quad Q_p = \frac{(P_C)_t^c \times (P_D)_t^d}{(P_A)_t^a \times (P_B)_t^b}$$

⊙ If reaction quotient (Q) < K_{eq} the reaction tends to move forward to attain eq^m.

⊙ If $Q > K_{eq}$, then reaction tends to move backward to attain eq^m.

⊙ If $Q = K_{eq}$, Equilibrium state

(2) Prediction of extent of reaction :

→ A very large value of K_{eq} (much larger than one) like 10^3, 10^4 etc. indicates that almost entire amount of reactants (at least limiting reactant) is completely converted into products.

→ A very low eqm. constant (much lower than 1) like 10^{-3}, 10^{-4}, 10^{-5} and lower indicate that reactants are only little converted into products.

(3.) To calculate change in concentration :

eg. $A_{(g)} + 2B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$, K_c or K_p

$t = 0$, a b c d moles initially

→ First calculate Q and compare with K_{eq}.

Here, Assume $Q < K_{eq}$

V = Volume

P_T = Total equilibrium pressure

$t = t_{eq}$, $(a-x)$ $(b-2x)$ $(c+x)$ $(d+x)$

$$K_c = \frac{[C][D]}{[A][B]^2} = \frac{\left(\frac{c+x}{V}\right)\left(\frac{d+x}{V}\right)}{\left(\frac{a-x}{V}\right)\left(\frac{b-2x}{V}\right)^2}$$

$$K_c = \frac{(c+x)(d+x)}{(a-x)(b-2x)^2} \times V$$

$$K_p = \frac{P_C \times P_D}{P_A \times P_B^2} = \frac{(X_C \times P_T)(X_D \times P_T)}{(X_A \times P_T)(X_B \times P_T)^2}$$

$$K_p = \frac{\left(\frac{c+x}{\Sigma n} \times P_T\right) \times \left(\frac{d+x}{\Sigma n} \times P_T\right)}{\left(\frac{a-x}{\Sigma n} \times P_T\right) \times \left(\frac{b+2x}{\Sigma n} \times P_T\right)^2}$$

$$K_p = \frac{(c+x)(d+x)}{(a-x)(b-2x)^2} \times \left(\frac{\Sigma n}{P_T}\right)$$

* Alternate Method of writing K_p & K_c :

eg. $\underset{(a-x)}{A} + \underset{(b-2x)}{2B} \rightleftharpoons \underset{(c+x)}{C} + \underset{(d+x)}{D}$, $\boxed{\Delta ng = -1}$

$\Delta ng = 2 - 3 = -1$

$$K_c = \frac{(c+x)(d+x)}{(a-x)(b-2x)} \times \left(\frac{1}{V}\right)^{\Delta ng}$$

$$\boxed{K_c = \frac{(c+x)(d+x)}{(a-x)(b-2x)} \times V}$$

$$K_p = \frac{(c+x)(d+x)}{(a-x)(b-2x)^2} \times \left(\frac{P_T}{\Sigma n}\right)^{\Delta ng}$$

$$\boxed{K_p = \frac{(c+x)(d+x)}{(a-x)(b-2x)^2} \times \frac{\Sigma n}{P_T}}$$

Q. 2 mol A and 3 mol of B are taken in a vessel. Find concentration of C at eq^m. if volume of vessel is 1 lit.

$$A(g) + B(g) \rightleftharpoons 2C(g), \quad K_c = 4 \text{ at } 100^\circ C$$

Soln

$$\underset{2}{A} + \underset{3}{2B} \rightleftharpoons \underset{0}{2C}, \quad Q_i = 0 < K_c$$

$$(2-x) \quad (3-x) \quad (2x)$$

$$K_c = 4 = \frac{(2x)^2}{(2-x)(3-x)} \times \left(\frac{1}{V}\right)^0$$

$$\Rightarrow 1 = \frac{x^2}{6-5x+x^2} \Rightarrow x^2 - 5x + 6 = x^2$$

$$\Rightarrow x = \frac{6}{5} = 1.2$$

$$[C]_{eq} = \frac{n_c}{V} = \frac{2x}{V} = \frac{2 \times 1.2}{1} = 2.4\ M$$

Q 1 mol of N_2 and 3 mol of H_2 are filled in a vessel of 1 lit capacity. Find conc. of NH_3 at eq^m. If

$$\underset{(g)}{N_2} + \underset{(g)}{3H_2} \rightleftharpoons 2NH_3(g), \quad K_c = \frac{4}{27} \text{ (mol/lit)}^{-2}$$

[Given $\sqrt{5} = 2.24$]

Soln.

$$\underset{1}{N_2(g)} + \underset{3}{3H_2} \rightleftharpoons \underset{0}{2NH_3}$$

$$1-x \quad 3-3x \quad 2x$$

$$K_c = \frac{4}{27} = \frac{(2x)^2}{(1-x)(3-3x)^3} \times \left(\frac{1}{V}\right)^{-2}$$

$$\frac{1}{16} = \frac{x^2}{(1-x)^4}$$

$$1 = \frac{x}{(1-x)^2} \Rightarrow 1 + x^2 - 2x - x = 0$$
$$\Rightarrow x^2 - 3x + 1 = 0$$

$$\Rightarrow x = \frac{+3 \pm \sqrt{9-4\times1}}{2} = \frac{+3 \pm 2.24}{2} = \frac{0.76}{2}, \frac{5.24}{2}$$

Now, $[NH_3]_{eq} = \frac{n_c}{V} = \frac{2x}{V} = \frac{0.76}{1} = 0.76$ mol/lit

Q. 3 mol of $PCl_5(g)$ is taken in 100 L vessel. Eqm pressure is 2.05 atm and temp. is 227°C. Calculate K_p for the reaction. (Take R = 0.082 lit atm $mol^{-1} K^{-1}$)

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

Soln

$PCl_5(g)$	$\rightleftharpoons$	$PCl_3(g)$	+	$Cl_2(g)$
3		0		0
(3-x)		x		x
1 mol		2 mol		2 mol

$$PV = nRT$$
$$2.05 \times 100 = (3 - x + x + x) \times 0.082 \times 300$$

$$(3+x) = \frac{2.05}{0.4} = 5$$

$$x = 2$$

$K_p = \frac{2 \times 2}{1} = 4 \times \left(\frac{2.05}{5}\right)^2$

$K_p = \frac{1}{5} \times 2.05 = 0.8 \times 2.05$

$K_p = 1.64$ atm.

(*) Relation between Vapour Density and 'α' (for a gaseous reaction)

Vapour density = $\frac{\text{Molecular weight}}{2}$

for a gas mixture

$(V.D)_{mixture} = \frac{M_{avg}}{2} = \left(\frac{\text{Total mass}}{\text{Total no. of moles}}\right) \times \frac{1}{2}$ ← Constant

$\Rightarrow V.D \propto \frac{1}{\text{Total no. of moles } (n_T)}$

$\frac{(V.D)_f}{(V.D)_i} = \frac{(n_T)_i}{(n_T)_f}$

a) For a dissociation reaction

$A(g) \rightleftharpoons nB(g)$, where $n > 1$

$t=0$,	a	0	where α = degree of dissociation of reactant
$t = t_{eq}$	$a(1-\alpha)$	$na\alpha$	

= $\frac{\text{No. of moles of Reactant Dissociated}}{\text{No. of moles of reactant initially taken}}$

$$(V.D) \propto \frac{1}{n_T}$$

$$\frac{(V.D)_i}{(V.D)_f} = \frac{(n_T)_f}{(n_T)_i}$$

$$\Rightarrow \frac{D}{d} = \frac{\phi(1-\alpha) + n\phi\alpha}{\phi}$$

$$\Rightarrow 1 + (n-1)\alpha = \frac{D}{d}$$

$$\Rightarrow \boxed{\alpha = \frac{(D-d)}{(n-1)d}} \quad \text{---} \quad (1)$$

$\Rightarrow$ D – Initial vapour Density
d – vapour density of eqm. mixture.
$(d < D)$ for dissociation.

$$\Rightarrow \text{Mavg} = \frac{\text{Total mass}}{n_T}$$

e.g. $2X_{(g)} \rightleftharpoons 3Y_{(g)} + 2Z_{(g)}$,
$n = \frac{5}{2}$

$$\Rightarrow (V.D)_{mix.} = \frac{\text{Mavg}}{2}$$

b) For association reaction : $(d > D)$

eg $\underset{a}{nX_{(g)}} \longrightarrow \underset{0}{Y_{(g)}}$ where $n > 1$

$a(1-\alpha)$. $\frac{a\alpha}{n}$ & α = degree of association

$$\frac{(V.D)_i}{(V.D)_f} = \frac{(n_T)_f}{(n_T)_i}$$

$$\Rightarrow \frac{D}{d} = \frac{d(1-\alpha) + \frac{\alpha d}{n}}{d}$$

$\Rightarrow$ α is independent of initial no. of moles of reactant.

$$\boxed{\alpha = \frac{(d-D)n}{d(n-1)}}$$

$$M_{avg} = \frac{\text{Total mass}}{\text{Total moles}}$$

eg. of association :

$$2NO_2(g) \underset{\text{cooling}}{\rightleftharpoons} N_2O_4(g)$$

Q. 2 mol of X (molar mass of X = 50 g/mol) dissociates as $2X(g) \rightleftharpoons 3Y(g) + 2Z(g)$

V.D at eqm. is found to be 16.

Find degree of dissociation of X.

Soln

	(2)	(0)	(0)
	$2X(g)$	$\rightleftharpoons 3Y(g)$	$+ 2Z(g)$
	$(2-2x)$	$3x$	$2x$

$$M_{avg} = 2 \times 16 = \frac{100}{(2-2x+3x+2x)} = \frac{50}{}$$

$$\Rightarrow (2+3x) = 25/8$$

$$\Rightarrow x = \frac{3}{8}$$

$$\therefore \alpha = \frac{2x}{2} = x = \frac{3}{8}$$

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Q1 Find % dissociation of $PCl_5(g)$ at 1 atm pressure.
$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, $K_p = 8$ atm.

Soln

$PCl_5(g)$	$\rightleftharpoons$	$PCl_3(g)$	+	$Cl_2(g)$
1		0		0
$(1-\alpha)$		α		α

$$K_p = 8 = \frac{(\alpha)\times(\alpha)}{(1-\alpha)} \times \left(\frac{1}{1+\alpha}\right)^1$$

$$\Rightarrow \quad 8 = \frac{\alpha^2}{1-\alpha^2}$$

$$\Rightarrow \quad \alpha = \frac{2\sqrt{2}}{3}$$

$$\therefore \text{ \% dissociation} = \frac{2\sqrt{2}}{3} \times 100 = \frac{2 \times 1.41 \times 100}{3}$$

$$\approx 94\% \quad \text{Ans}$$

***** Approximation

Q2 Find eq^m. concentration of each gas in 10 lit. vessel if 1 mol of PCl_5 is filled to attain eq^m as ~~$PCl_5(g)$~~
$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, $K_c = 10^{-5}$ M

Soln

$PCl_5(g)$	$\rightleftharpoons$	$PCl_3(g)$	+	$Cl_2(g)$
1		0		0
$(1-x)$		x		x

$$K_c = 10^{-5} = \frac{x \times x}{(1-x)} \times \left(\frac{1}{10}\right)^1$$

$$10^{-4} = \frac{x^2}{1-x}$$

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Since, K_c is very small so, $x <<<< 1$ ∴ $(1-x) \approx 1$.

$$x^2 \approx 10^{-4}$$
$$x \approx 10^{-2}\ M$$

$$[PCl_3] = [Cl_2] = \frac{x}{V} = \frac{10^{-2}}{10} = 10^{-3}\ M$$

$$[PCl_5] = \frac{1-x}{10} = 0.1\ M.$$

imp Q. In above qn. if K_c is 10^4, find conc. of each gas.

Soln

$PCl_5(g)$	$\rightleftharpoons$	$PCl_3(g)$ +	$Cl_2(g)$
1		0	0
(1-x)		1	1

$$\Rightarrow K_c = \frac{1 \times 1}{x} \times \left(\frac{1}{10}\right)^1 = 10^4$$

$$\Rightarrow x = 10^{-5} \text{ moles}$$

$$[PCl_3] = [Cl_2] = \frac{1}{10} = 0.1\ M$$

$$[PCl_5]_f = \frac{x}{V} = \frac{10^{-5}}{10} = 10^{-6}\ M$$

* Le - Chatelier Principle :
(Effect of conc., Pressure, temperature, inert gas etc) on equilibrium System.

→ If one of the parameter like conc., pressure, temperature etc of an equilibrium system is changed then equilibrium moves in the direction such that the effect of caused change in nullified as much as possible.

→ However, note that the changed parameter need not come back to its original value.
when eq^m. is re-established.

(1.) Effect of Concentration (or partial pressure) Variation :-

eg. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, $K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]}$

Concentration at eq. → a, b, c

$(a+x)$ b c (Not at eq^m)

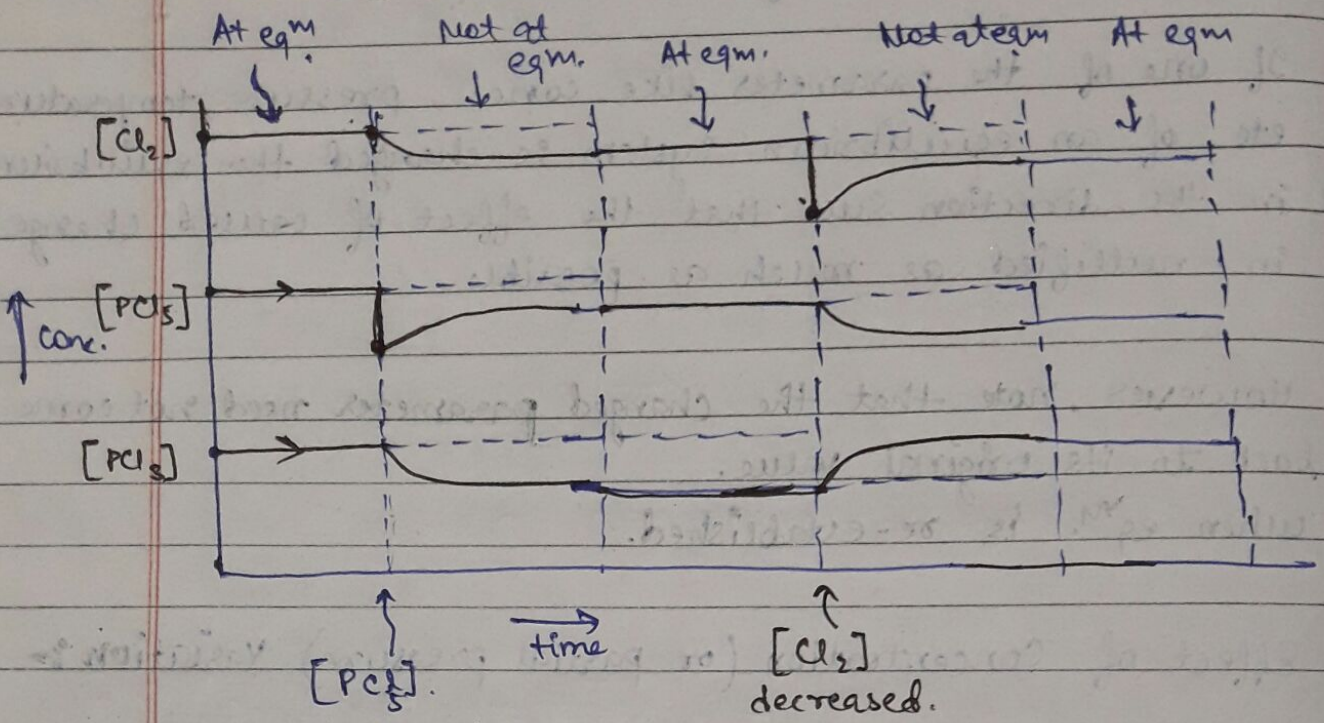
a' b' c' (At new eq^m)

$\Rightarrow a < a' < (a+x)$ | $b < b'$ | $c < c'$

If conc. of any reactant is increased for an eq^m system then eq^m. shifts in forward direction.

Graphically —

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$



(2) Effect of System pressure variation (or Effect of volume variation)

(a) For an eqm. system involving gas (gases) with or without pure solids / liquids.

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, $K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]}$

$P_{sys.} \uparrow$ (or $V \downarrow$), Backward shift

$Q_c > K_c$
↳ Eqm. shifts backward

⁂ On increase in system pressure (by decreasing volume of system), Eqm. shifts (for gaseous reaction) in the direction which produces lesser no. of gas moles.

In above case :-

Eqm. shifted backward

$n_{PCl_5} \uparrow$, $n_{PCl_3} \downarrow$, $n_{Cl_2} \downarrow$

But, Conc. or partial pressure of PCl_5, PCl_3, Cl_2 all are increased

$$Conc = \frac{n}{(V)}$$ → dominant Effect

eg. $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$; $\Delta n_g = 0$

$(P_{system}) \uparrow$ or $(V \downarrow)$, Eqm is not disturbed.

⇒ All conc. or partial pressure are increased.

(b) For a reaction or phase change involving only solids and or liquids :

→ for an eqm. system involving only solids and liquids if system pressure is increased then eqm. shifts in the direction which produces more dense phase.

→ In general, $d_{solid} > d_{liquid}$ (for a substance)

Exception : $d_{ice} < d_{H_2O(l)}$

eg $S(s) \rightleftharpoons S(l)$, ΔH = +ve (Endothermic)

$(P_{sys})\uparrow$, Eq^m. shifts backward, M.P increases.

eg $H_2O(s) \rightleftharpoons H_2O(l)$

$(P_{sys})\uparrow$, Eq^m. shifts forward; M.P decreases.

eg. $C_{(graphite)} \rightleftharpoons C_{(diamond)}$ (↓ more density)

$(P_{system})\uparrow$, Eq^m. shifts forward.

(3) Effect of introduction of inert gas :

(a) At constant volume and temperature.

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, $K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]}$

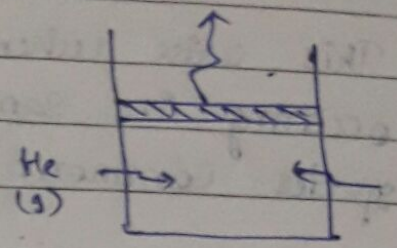
→ If An inert gas is introduced isothermally and isochorically, then eq^m. is undisturbed. All the conc. or partial pressure of eq^m. system remain unaffected.

$[Cl_2] = \frac{n_{Cl_2}}{V}$ → Constant, V → Constant

→ However, total pressure of system will increase.

(b) At constant pressure and temperature.

$PV = nRT$



$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

V → increases.

$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]}$ ~~घटी × घटी / घटी~~

$Q_c < K_c$,

Eqm. shifts forward.

→ If an inert gas is introduced isobarically (const. ~~temperature~~ pressure) and isothermally (const. temperature)

then, eqm. shifts in the direction which produces greater no. of gas moles.

→ Due to volume increase, at new eqm. all conc. or partial pressure will decrease.

(4) Effect of temperature :

→ On increase in temperature, Eqm. shifts forward for endothermic (ΔH = +ve) and shifts backward for exothermic rxn. (ΔH = -ve)

(5) Effect of Catalyst :
No effect of catalyst on eqm.

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* Simultaneous Equilibrium :

→ This arise when two or more equilibria are occuring in same system with atleast one of the species common.

$$\underset{P_1'}{X_{(s)}} \rightleftharpoons \underset{P_1'}{A_{(g)}} + B_{(g)} \quad ; \quad (K_{P_1}) = P_1'^2$$

$$Y_{(s)} \rightleftharpoons \underset{P_2'}{A_{(g)}} + \underset{P_2'}{C_{(g)}} \quad ; \quad (K_{P_2}) = P_2'^2$$

⇒ Now simultaneously X and Y solids are taken in vessel and allowed to dissociate.

$$X_{(s)} \rightleftharpoons \underset{(P_1+P_2)}{A_{(g)}} + \underset{P_1}{B_{(g)}} \quad ; \quad K_{P_1} = P_1'^2$$

$$Y_{(s)} \rightleftharpoons \underset{(P_2+P_1)}{A_{(g)}} + \underset{P_2}{C_{(g)}} \quad ; \quad K_{P_2} = P_2'^2$$

⇒ $P_1 < P_1'$

⇒ $P_2 < P_2'$

$$K_{P_1} = P_1'^2 = (P_1 + P_2) \times P_1 \longrightarrow (i)$$

$$K_{P_2} = P_2'^2 = (P_1 + P_2) \times P_2 \longrightarrow (ii)$$

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Q. Pressure over solid (X) is found to be 20 torr and x dissociates as

$$X(s) \rightleftharpoons A(g) + B(g)$$

Pressure over solid 'Y' is found to be 40 torr and Y dissociates as

$$Y(s) \rightleftharpoons A(g) + C(g)$$

Find (i) K_p of both the reactions.
(ii) total pressure at eqm. when X and Y are taken in enough amount simultaneously.

Soln (i) $X(s) \rightleftharpoons A(g) + B(g)$, $K_{P_1} = 10^2 = 100\ torr^2$

~~10 torr~~ ~~10 torr~$

$(P_1 + P_2)$ P_1

$Y(s) \rightleftharpoons A(g) + C(g)$, $K_{P_2} = 20^2\ torr^2 = 400\ torr^2$

~~20 torr~~ ~~20 torr~~

$(P_2 + P_1)$ P_2

(ii)

$$K_{P_1} = 100 = (P_1 + P_2) \times P_1 \quad \text{---(i)}$$
$$K_{P_2} = 400 = (P_2 + P_1) \times P_2 \quad \text{---(ii)}$$

$$(P_1 + P_2)^2 = 500$$
$$(P_1 + P_2) = 10\sqrt{5}\ torr$$

$$(P_T)_{eq} = P_A + P_B + P_C$$
$$= (P_1 + P_2) + P_1 + P_2$$
$$= 2(P_1 + P_2)$$
$$= 20\sqrt{5}\ torr$$

Q. Following two eqm. are established on mixing two gases A_2 and C.

$$3A_2(g) \rightleftharpoons A_6(g), \quad K_p = 1.6 \text{ atm}^{-2}$$
$$A_2(g) + C(g) \rightleftharpoons A_2C(g), \quad K_p = x$$

If A_2 and C are initially mixed in 2:1 molar ratio, the total pressure and partial pressure of A_6 are 1.4 atm and 0.2 atm respectively.

Find x.

Soln $3A_2(g) \rightleftharpoons A_6(g)$, $K_p = 1.6 \Rightarrow x = 0.6$ atm.

(2p)-x-y) x/3

$A_2(g) + C(g) \rightleftharpoons A_2C(g)$, $K_p = ?$

(2p-x-y) (p)-y y

$K_p \Rightarrow 1.6 = \dfrac{0.2}{(2p-x-y)^3} \Rightarrow (2p-x-y) = 0.5$

$\Rightarrow (2p-y) = 1.1$

$2p - 0.6 - y + 0.2 + p = 1.4$

$3p - y = 1.8$ —— (ii)

$P = 0.7$ atm, $y = 2.1 - 1.8 = 0.3$ atm

$K_p = x = \dfrac{0.3}{(1.4-0.6-0.3)\times(0.7-0.3)}$

$= x = \dfrac{0.3}{0.5 \times 0.4} = 1.5 \text{ atm}^{-1}$

* Free Energy and Equilibrium :

Reversible Reaction :

$$aA(g) + bB(g) \rightleftharpoons cC(g) + dD(g)$$

At any instant,

$$\boxed{\Delta G_r = \Delta G_r^\circ + 2.303RT \log_{10} Q_p}$$

ΔG_r = Gibb's Energy of reaction.
ΔG_r° = Standard Gibb's Energy of Reaction.
T = Temperature in Kelvin
Q_p = Reaction quotient.

⇒ Standard conditions :
When gases → ideal
Partial Pressure = 1 bar.

→ For substance in solution phase : Conc = 1 Molar
→ For ~~pure~~ solid / liquid = Pure and at 1 bar pressure.

* Temperature is not included under Standard conditions.

$$\Delta G_r = \Delta G_r^\circ + 2.303RT \log_{10} Q_p \quad \text{— (i)}$$

At equilibrium state,
⇒ $\Delta G_r = 0$, $Q_p = K_p$

⇒ $0 = \Delta G_r^\circ + 2.303RT \log_{10} Q_p$

⇒ $\boxed{\Delta G_r^\circ = -2.303RT \log_{10} K_p}$ — (ii)

From thermodynamics,

$$\Delta G_r^\circ = \Delta H_r^\circ - T\Delta S_r^\circ \quad \text{(iii)}$$

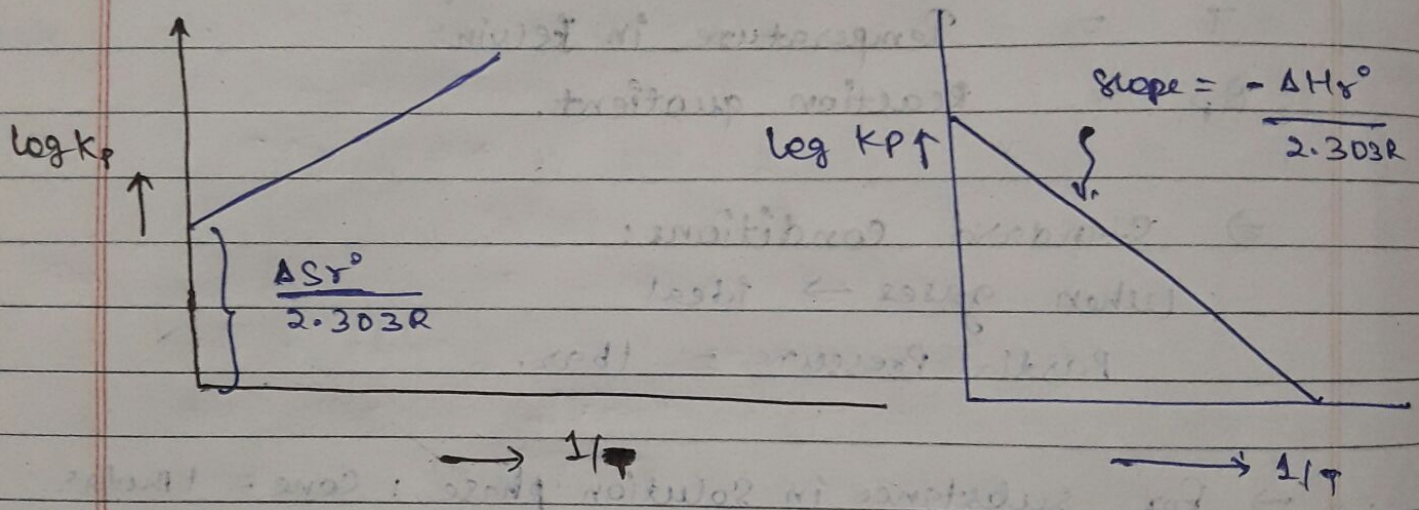
ΔH_r° - Standard enthalpy of reaction

ΔS_r° - Standard entropy of reaction.

$$\Rightarrow -2.303RT \log K_p = \Delta G_r^\circ = \Delta H_r^\circ - T\Delta S_r^\circ$$

$$\Rightarrow \log K_p = \frac{-\Delta H_r^\circ}{2.303RT} + \frac{\Delta S_r^\circ}{2.303R}$$

$y = mx + c$



For exothermic rkn
ΔH_r° = -ve

For endothermic rkn
ΔH_r° = +ve

$\Rightarrow$ For a reaction at eq^m. $\Delta G_r = 0$ and ΔG_r° need not be equal to zero.

$\Rightarrow$ $\Delta G_r^\circ = 0$ for a reaction at eq^m if all partial pressure of gases at eq^m. = 1 bar

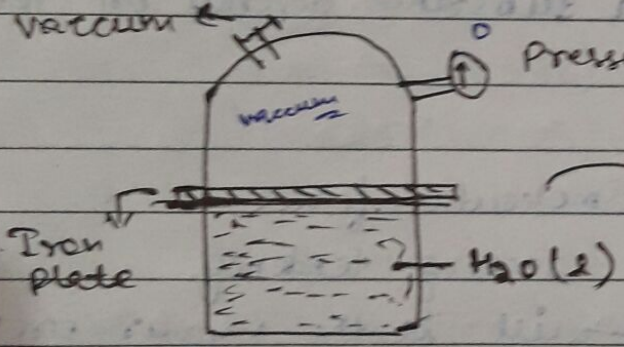
$$\Delta G_r^\circ = -2.303RT \log K_p$$
$$0 = -2.303RT \log K_p \Rightarrow K_p = 1$$

⇒ $\Delta G_r^\circ = -2.303 RT \log_{10} K_p$

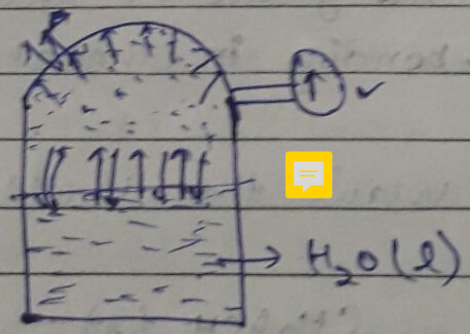
If ΔG_r = +ve, then reaction is non-spontaneous
If ΔG_r = -ve, then rxn is spontaneous and moves in forward direction.

(A) Vapour Pressure :-

Vapour pressure of a substance is defined as the pressure exerted by the vapours of a substance (solid or liquid) in eqm. state.



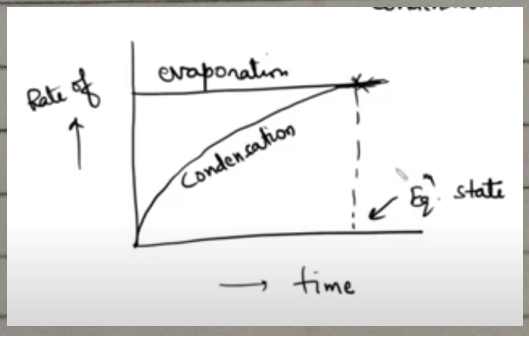
Iron plate is removed



At dynamic eqm. state

Rate of evaporation = Rate of condensation.

$$H_2O(l) \underset{\text{condensation}}{\overset{\text{evaporation}}{\rightleftharpoons}} H_2O(g)$$



→ Most liquids are volatile.
$Hg(l)$ → non volatile under room condition

→ Few solids are volatile
eg. $I_2(s)$ is volatile

Note :

(1.) Vapour pressure of a substance does not depend on
(i) Surface Area.
(ii) Amount of substance

(2) Vapour pressure of a substance depends on temperature
T↑, V.P↑

(3) Stronger are the intermolecular forces,
lesser will be the escaping tendency.
and therefore lower will be the vapour pressure

eg. Ethanol (C_2H_5OH) has greater V.P than water, due to weaker H-bonding in ethanol than water.

Or Ethanol is more volatile than $H_2O(l)$

eg. $CH_3CH_2OH(l)$ $CH_3OCH_3(l)$

↑ Ether has higher V.P than $C_2H_5OH(l)$ (lack of H-bonding)

⇒ Zero V.P ⇒ Non-volatile substance

⇒ V.P ↑ Boiling Point ↓

④ Relative Humidity (R.H)

$$= \frac{\text{Partial Pressure of } H_2O(g)}{\text{V.P of } H_2O} \times 100$$

* Both these pressure are measured at same temperature.

* V.P of $H_2O(l)$ is also called as aqueous tension.

At equilibrium of $H_2O(l) \rightleftharpoons H_2O(g)$

$P_{H_2O(g)}$ = V.P of H_2O

Then RH = 100%.

In such case Air/gas is said to be saturated w.r.t moisture.

* **Hygroscopic substances: Substances which absorb water, but not enough to dissolve in it ex. CaO, NaNO3, sucrose and CuO**

* Deliquescent substance: / Hygroscopic sub.

Substance which absorb moisture and become wet when exposed to atmospheric air.

eg. NaOH, $MgCl_2$, Na_2SO_4 etc.

* Efflorescent substance:

Substance which loose moisture when exposed to atmospheric air

eg. Potash Alum ($K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$)

Glauber's salt ($Na_2SO_4 \cdot 10H_2O$)

Imp

Q $Na_2SO_4 \cdot 10H_2O(s) \rightleftharpoons Na_2SO_4(s) + 10H_2O(g)$

$(K_p)_{0^\circ C} = 1.024 \times 10^{-27}$ atm^{10}

(i) At what R.H will $Na_2SO_4 \cdot 10H_2O$ be efflorescent when exposed to air at 0°C

(ii) At what RH will Na_2SO_4 be deliquescent when exposed to air at 0°C.

Given

$(V.P)_{H_2O(l), 0^\circ C} = 4.56$ torr

Soln. $K_p = 1.024 \times 10^{-27} = (P_{H_2O})^{10}_{eq} \Rightarrow (P_{H_2O})_{eq} = 2 \times 10^{-3}$ atm

$(P_{H_2O})_{eq} = 0.002 \times 760$

$= 1.52$ torr

$(RH)_{eqm.} = \frac{1.52}{4.56} \times 100 = 33.3\%$

$Na_2SO_4 \cdot 10H_2O(s) \rightleftharpoons Na_2SO_4(s) + 10H_2O(g)$, K_p

(i) When $Q_p < K_p$, $Na_2SO_4 \cdot 10H_2O$ will be efflorescent

i.e RH should be less than 33.3%

(ii) Na_2SO_4 will be deliquescent when RH > 33.3%

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Q. Solid ammonium carbonate, NH_2COONH_4 dissociates as $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

At eqm. total pressure is observed to be 6 atm. Find K_p of rxn.

Soln. $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

$(2p) \quad (p)$

$2p + p = 6 \Rightarrow p = 2 \text{ atm}$

$K_p = (2p)^2 \times (p) = 4p^3 = 4 \times 2^3 = 32 \text{ atm}^3$